

# Daily Problem Solutions Booklet

*Comprehensive Step-by-Step Analytical Solutions*

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## Integrals

### 1.1 Medium Integral

#### Symmetric Trigonometric Exponent Integral

Evaluate the definite integral:

$$\int_0^{\pi/2} \frac{\sin(x)^{\cos(x)}}{\sin(x)^{\cos(x)} + \cos(x)^{\sin(x)}} dx$$

**Solution:** Let the given integral be denoted as  $I$ :

$$I = \int_0^{\pi/2} \frac{\sin(x)^{\cos(x)}}{\sin(x)^{\cos(x)} + \cos(x)^{\sin(x)}} dx$$

We utilize King's property for definite integrals over the interval  $[0, a]$ , which states that:

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

Applying this property with  $a = \frac{\pi}{2}$ , we replace every instance of  $x$  with  $(\frac{\pi}{2} - x)$ . Using the standard complementary angle identities  $\sin(\frac{\pi}{2} - x) = \cos(x)$  and  $\cos(\frac{\pi}{2} - x) = \sin(x)$ , we obtain:

$$I = \int_0^{\pi/2} \frac{\sin(\frac{\pi}{2} - x)^{\cos(\frac{\pi}{2} - x)}}{\sin(\frac{\pi}{2} - x)^{\cos(\frac{\pi}{2} - x)} + \cos(\frac{\pi}{2} - x)^{\sin(\frac{\pi}{2} - x)}} dx = \int_0^{\pi/2} \frac{\cos(x)^{\sin(x)}}{\cos(x)^{\sin(x)} + \sin(x)^{\cos(x)}} dx$$

Notice that both representations of  $I$  share the exact same denominator. Adding the two equations together yields:

$$2I = \int_0^{\pi/2} \frac{\sin(x)^{\cos(x)} + \cos(x)^{\sin(x)}}{\sin(x)^{\cos(x)} + \cos(x)^{\sin(x)}} dx$$

The integrand simplifies completely to unity:

$$2I = \int_0^{\pi/2} 1 dx = [x]_0^{\pi/2} = \frac{\pi}{2}$$

Dividing both sides by 2 gives the final exact value:

$$I = \frac{\pi}{4}$$

### 1.2 Hard Integral

### Gamma Function Product Integral

Evaluate the integral:

$$\int_0^\infty x \Gamma^2 \left( 1 + \frac{ix^2}{\pi} \right) \Gamma^2 \left( 1 - \frac{ix^2}{\pi} \right) dx$$

**Solution:** Let the integral be  $I$ . We first simplify the product of Gamma functions using Euler's reflection formula,  $\Gamma(1+z)\Gamma(1-z) = \frac{\pi z}{\sin(\pi z)}$ . Setting  $z = \frac{ix^2}{\pi}$ , we get:

$$\Gamma \left( 1 + \frac{ix^2}{\pi} \right) \Gamma \left( 1 - \frac{ix^2}{\pi} \right) = \frac{\pi \left( \frac{ix^2}{\pi} \right)}{\sin \left( \pi \frac{ix^2}{\pi} \right)} = \frac{ix^2}{\sin(ix^2)}$$

Using the hyperbolic identity  $\sin(i\theta) = i \sinh(\theta)$ , this expression simplifies to:

$$\frac{ix^2}{i \sinh(x^2)} = \frac{x^2}{\sinh(x^2)}$$

Squaring this product gives:

$$\Gamma^2 \left( 1 + \frac{ix^2}{\pi} \right) \Gamma^2 \left( 1 - \frac{ix^2}{\pi} \right) = \frac{x^4}{\sinh^2(x^2)}$$

Substitute this back into the integral:

$$I = \int_0^\infty x \cdot \frac{x^4}{\sinh^2(x^2)} dx = \int_0^\infty \frac{x^5}{\sinh^2(x^2)} dx$$

Apply the substitution  $u = x^2$ , which implies  $du = 2x dx$ , or  $x dx = \frac{1}{2} du$ . The integral transforms into:

$$I = \frac{1}{2} \int_0^\infty \frac{u^2}{\sinh^2(u)} du$$

To evaluate this integral, we apply integration by parts. Let  $f(u) = u^2 \implies f'(u) = 2u$ , and  $g'(u) = \operatorname{csch}^2(u) \implies g(u) = -\coth(u) + 1 = \frac{2e^{-2u}}{1-e^{-2u}}$  (choosing the integration constant to ensure convergence at infinity):

$$\int_0^\infty u^2 \operatorname{csch}^2(u) du = \left[ -u^2(\coth(u) - 1) \right]_0^\infty + 2 \int_0^\infty u(\coth(u) - 1) du$$

The boundary term evaluates to zero at both limits. At infinity,  $(\coth(u) - 1) \sim 2e^{-2u}$ , which decays exponentially faster than  $u^2$  grows. At zero,  $u^2 \coth(u) \sim u \rightarrow 0$ . Thus:

$$\int_0^\infty \frac{u^2}{\sinh^2(u)} du = 2 \int_0^\infty u \left( \frac{2e^{-2u}}{1-e^{-2u}} \right) du = 4 \int_0^\infty u \sum_{n=1}^\infty e^{-2nu} du$$

Swapping the order of integration and summation:

$$4 \sum_{n=1}^\infty \int_0^\infty u e^{-2nu} du = 4 \sum_{n=1}^\infty \frac{1}{(2n)^2} = \sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$$

Substituting this result back into our expression for  $I$ :

$$I = \frac{1}{2} \left( \frac{\pi^2}{6} \right) = \frac{\pi^2}{12}$$

## Differentials

### 2.1 Medium Differential

#### Non-Linear Volterra Integral Equation

Compute  $f^{-1}(2)$  for all  $x \geq 0$  if:

$$f(x) = 1 + \int_0^x \frac{t}{1 + f(t)} dt$$

**Solution:** First, differentiate both sides of the integral equation with respect to  $x$  using the Fundamental Theorem of Calculus:

$$f'(x) = \frac{x}{1 + f(x)}$$

From the original integral equation, evaluating at  $x = 0$  gives the initial condition:

$$f(0) = 1 + \int_0^0 \frac{t}{1 + f(t)} dt = 1 + 0 = 1$$

We now solve the separable differential equation by writing  $y = f(x)$ :

$$(1 + y) dy = x dx$$

Integrating both sides yields:

$$\int (1 + y) dy = \int x dx \implies y + \frac{y^2}{2} = \frac{x^2}{2} + C$$

Multiply the entire equation by 2 to clear denominators:

$$y^2 + 2y = x^2 + 2C$$

Complete the square on the left-hand side:

$$(y + 1)^2 - 1 = x^2 + 2C \implies (y + 1)^2 = x^2 + K$$

Using the initial condition  $y(0) = 1$  at  $x = 0$  to find the constant  $K$ :

$$(1 + 1)^2 = 0^2 + K \implies K = 4$$

The algebraic relation defining the function is therefore:

$$(f(x) + 1)^2 = x^2 + 4$$

Since  $f(0) = 1 > -1$ , we take the positive square root:

$$f(x) + 1 = \sqrt{x^2 + 4} \implies f(x) = \sqrt{x^2 + 4} - 1$$

We are required to compute  $f^{-1}(2)$ , which means finding the value of  $x \geq 0$  such that  $f(x) = 2$ :

$$\sqrt{x^2 + 4} - 1 = 2 \implies \sqrt{x^2 + 4} = 3$$

Square both sides:

$$x^2 + 4 = 9 \implies x^2 = 5$$

Since the domain is specified as  $x \geq 0$ , we take the principal root:

$$f^{-1}(2) = \sqrt{5}$$

## 2.2 Hard Differential

### Euler's Numerical Approximation

Consider the differential equation:

$$\frac{dy}{dx} = \frac{1}{3 + 3x + x^2}$$

Given the initial condition  $y(0) = 1$ , use Euler's method with a step size of  $h = 0.1$  to find an approximation for  $y(0.2)$ . If the approximation is expressed as a coprime fraction  $\frac{a}{b}$ , determine the value of the denominator  $b$ .

**Solution:** Euler's numerical method updates the approximation using the recursive formula:

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

where  $f(x, y) = \frac{1}{x^2 + 3x + 3}$ , step size  $h = 0.1 = \frac{1}{10}$ , and initial values  $x_0 = 0, y_0 = 1$ .

**Step 1: Compute  $y_1$  at  $x_1 = 0.1$**  Evaluate the slope at  $(x_0, y_0)$ :

$$f(x_0, y_0) = f(0, 1) = \frac{1}{0^2 + 3(0) + 3} = \frac{1}{3}$$

Apply Euler's formula to find  $y_1$ :

$$y_1 = y_0 + h \cdot f(0, 1) = 1 + \frac{1}{10} \times \frac{1}{3} = 1 + \frac{1}{30} = \frac{31}{30}$$

**Step 2: Compute  $y_2$  at  $x_2 = 0.2$**  Evaluate the slope at  $(x_1, y_1) = \left(\frac{1}{10}, \frac{31}{30}\right)$ :

$$f(x_1, y_1) = \frac{1}{\left(\frac{1}{10}\right)^2 + 3\left(\frac{1}{10}\right) + 3} = \frac{1}{\frac{1}{100} + \frac{3}{10} + 3} = \frac{1}{\frac{1+30+300}{100}} = \frac{100}{331}$$

Apply Euler's formula to find  $y_2$ :

$$y_2 = y_1 + h \cdot f(x_1, y_1) = \frac{31}{30} + \frac{1}{10} \times \frac{100}{331} = \frac{31}{30} + \frac{10}{331}$$

Combine the two fractions over a common denominator:

$$y_2 = \frac{31 \times 331 + 30 \times 10}{30 \times 331} = \frac{10261 + 300}{9930} = \frac{10561}{9930}$$

We now check if the fraction  $\frac{10561}{9930}$  is in simplest coprime form. The prime factorization of the denominator is:

$$9930 = 10 \times 993 = 2 \times 5 \times 3 \times 331$$

Testing divisibility of the numerator 10561:

- It is odd and does not end in 0 or 5, so it is not divisible by 2 or 5.

- The sum of digits is  $1 + 0 + 5 + 6 + 1 = 13$ , which is not divisible by 3.
- Dividing by 331 yields  $10561 = 31 \times 331 + 300$ , leaving a remainder of 300, so it is not divisible by 331.

Thus,  $\gcd(10561, 9930) = 1$ , confirming the fraction is fully reduced. The required value of the denominator  $b$  is:

$$b = 9930$$

## Derivatives

### 3.1 Medium Derivative

#### Anger Function Higher-Order Derivative

The Anger Function is defined as:

$$J_u(x) = \frac{1}{\pi} \int_0^\pi \cos(u\theta - x \sin(\theta)) d\theta$$

Compute  $J_0''''(0)$ .

**Solution:** For order  $u = 0$ , the Anger function simplifies to:

$$J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(-x \sin(\theta)) d\theta = \frac{1}{\pi} \int_0^\pi \cos(x \sin(\theta)) d\theta$$

This is identical to the integral representation of the zero-order Bessel function of the first kind,  $J_0(x)$ . We differentiate under the integral sign with respect to  $x$ . Applying the chain rule successively:

$$\frac{d}{dx} \cos(x \sin(\theta)) = -\sin(\theta) \sin(x \sin(\theta))$$

$$\frac{d^2}{dx^2} \cos(x \sin(\theta)) = -\sin^2(\theta) \cos(x \sin(\theta))$$

$$\frac{d^3}{dx^3} \cos(x \sin(\theta)) = \sin^3(\theta) \sin(x \sin(\theta))$$

$$\frac{d^4}{dx^4} \cos(x \sin(\theta)) = \sin^4(\theta) \cos(x \sin(\theta))$$

Evaluating this fourth derivative at  $x = 0$ :

$$\left. \frac{d^4}{dx^4} \cos(x \sin(\theta)) \right|_{x=0} = \sin^4(\theta) \cos(0) = \sin^4(\theta)$$

Therefore, the fourth derivative of the Anger function at zero is:

$$J_0''''(0) = \frac{1}{\pi} \int_0^\pi \sin^4(\theta) d\theta$$

We evaluate this trigonometric integral using power reduction identities:

$$\sin^4(\theta) = \left( \frac{1 - \cos(2\theta)}{2} \right)^2 = \frac{1 - 2\cos(2\theta) + \cos^2(2\theta)}{4} = \frac{1 - 2\cos(2\theta) + \frac{1 + \cos(4\theta)}{2}}{4}$$

$$\sin^4(\theta) = \frac{3 - 4\cos(2\theta) + \cos(4\theta)}{8}$$



Integrating term-by-term from 0 to  $\pi$ :

$$\int_0^\pi \sin^4(\theta) d\theta = \int_0^\pi \frac{3}{8} d\theta - \int_0^\pi \frac{\cos(2\theta)}{2} d\theta + \int_0^\pi \frac{\cos(4\theta)}{8} d\theta$$

Since the cosine terms integrate to zero over full periods:

$$\int_0^\pi \sin^4(\theta) d\theta = \frac{3\pi}{8} - 0 + 0 = \frac{3\pi}{8}$$

Multiplying by the prefactor  $\frac{1}{\pi}$  yields:

$$J_0'''(0) = \frac{1}{\pi} \left( \frac{3\pi}{8} \right) = \frac{3}{8}$$

### 3.2 Hard Derivative

#### Infinite Rational Series Derivative Symmetry

Given the function:

$$f(x) = \sum_{n=1}^{\infty} \frac{x + n(-1)^n}{x^2 + n^2}$$

Find  $f'(-1) + f'(1)$ .

**Solution:** We split the function  $f(x)$  into two distinct sums:

$$f(x) = u(x) + v(x) = \sum_{n=1}^{\infty} \frac{x}{x^2 + n^2} + \sum_{n=1}^{\infty} \frac{n(-1)^n}{x^2 + n^2}$$

Let us analyze the parity (symmetry) of each component function:

- $u(-x) = \sum_{n=1}^{\infty} \frac{-x}{(-x)^2 + n^2} = -u(x)$ , so  $u(x)$  is an **odd** function. Its derivative  $u'(x)$  is an **even** function, meaning  $u'(-1) = u'(1)$ .
- $v(-x) = \sum_{n=1}^{\infty} \frac{n(-1)^n}{(-x)^2 + n^2} = v(x)$ , so  $v(x)$  is an **even** function. Its derivative  $v'(x)$  is an **odd** function, meaning  $v'(-1) = -v'(1)$ .

Using these symmetry properties, we evaluate the required sum:

$$f'(-1) + f'(1) = [u'(-1) + v'(-1)] + [u'(1) + v'(1)] = [u'(1) - v'(1)] + [u'(1) + v'(1)] = 2u'(1)$$

Notice that the alternating terms  $v'(x)$  cancel out completely! We only need to compute  $2u'(1)$ . From the well-known Mittag-Leffler pole expansion of the hyperbolic cotangent function:

$$\pi \coth(\pi x) = \frac{1}{x} + \sum_{n=1}^{\infty} \frac{2x}{x^2 + n^2} = \frac{1}{x} + 2u(x)$$

Isolating  $2u(x)$ :

$$2u(x) = \pi \coth(\pi x) - \frac{1}{x}$$

Differentiating both sides with respect to  $x$ :

$$2u'(x) = \frac{d}{dx} \left( \pi \coth(\pi x) - \frac{1}{x} \right) = -\pi^2 \operatorname{csch}^2(\pi x) + \frac{1}{x^2}$$

Evaluating this derivative at  $x = 1$ :

$$f'(-1) + f'(1) = 2u'(1) = -\pi^2 \operatorname{csch}^2(\pi) + \frac{1}{1^2} = 1 - \frac{\pi^2}{\sinh^2(\pi)}$$

## Physics & Engineering

### 4.1 Mechanics

#### Rolling Without Slipping Dynamics

A cylinder with mass  $m = 1 \text{ kg}$  and radius  $R = 0.1 \text{ m}$  moves at  $v_0 = 10 \text{ m s}^{-1}$  on a frictionless plane. At  $t = 0$  it enters a region with friction coefficient  $\mu = 0.2$ . A string at the top is pulled with a force  $T = 1 \text{ N}$  to the left. Find the time  $t$  (in s) until the cylinder rolls without slipping. Take  $g = 9.8 \text{ m s}^{-2}$ .

**Solution:** Let the positive horizontal axis ( $+x$ ) be directed to the right, and positive angular displacement ( $\theta$ ) be clockwise. Initially at  $t = 0$ , the cylinder moves with purely translational velocity  $v_0 = 10 \text{ m s}^{-1}$  to the right without rotation ( $\omega_0 = 0$ ). Upon entering the rough region, the bottom contact point has a rightward velocity  $v_{\text{bottom}} = v_0 - \omega_0 R = 10 \text{ m s}^{-1} > 0$ . Therefore, kinetic friction  $f_k$  opposes this motion and acts to the left:

$$f_k = \mu N = \mu mg = 0.2 \times 1 \text{ kg} \times 9.8 \text{ m s}^{-2} = 1.96 \text{ N}$$

Two forces act horizontally: the tension  $T = 1 \text{ N}$  (to the left) at the top, and kinetic friction  $f_k = 1.96 \text{ N}$  (to the left) at the bottom. By Newton's Second Law for translation:

$$F_{\text{net}} = -T - f_k = -1 - 1.96 = -2.96 \text{ N}$$

$$a = \frac{F_{\text{net}}}{m} = \frac{-2.96 \text{ N}}{1 \text{ kg}} = -2.96 \text{ m s}^{-2}$$

The linear velocity as a function of time is:

$$v(t) = v_0 + at = 10 - 2.96t$$

Next, analyze the net torque about the center of mass (with clockwise taken as positive):

- Friction  $f_k$  at the bottom ( $y = -R$ ) acts to the left, creating a clockwise (positive) torque:  $\tau_{f_k} = +f_k R$ .
- Tension  $T$  at the top ( $y = +R$ ) acts to the left, creating a counter-clockwise (negative) torque:  $\tau_T = -TR$ .

$$\tau_{\text{net}} = (f_k - T)R = (1.96 - 1) \times 0.1 = 0.96 \times 0.1 = 0.096 \text{ N m}$$

For a uniform solid cylinder, the moment of inertia about its center of mass is:

$$I = \frac{1}{2}mR^2 = \frac{1}{2}(1 \text{ kg})(0.1 \text{ m})^2 = 0.005 \text{ kg m}^2$$

The angular acceleration  $\alpha$  is:

$$\alpha = \frac{\tau_{\text{net}}}{I} = \frac{0.096 \text{ N m}}{0.005 \text{ kg m}^2} = 19.2 \text{ rad s}^{-2}$$

The angular velocity as a function of time is:

$$\omega(t) = \omega_0 + \alpha t = 0 + 19.2t = 19.2t$$

The condition for rolling without slipping is reached when the relative velocity of the bottom contact point becomes zero ( $v_{\text{bottom}} = v(t) - \omega(t)R = 0$ ), meaning:

$$v(t) = \omega(t)R$$

Substitute the derived kinematic equations into this condition:

$$10 - 2.96t = (19.2t)(0.1)$$

$$10 - 2.96t = 1.92t \implies 4.88t = 10$$

Solving for the time  $t$ :

$$t = \frac{10}{4.88} = \frac{125}{61} \approx 2.049 \text{ s}$$

## 4.2 Quantum Mechanics

### Synchrotron Radiation Energy Loss

An electron travels in a synchrotron with an energy of  $E = 7.50 \text{ GeV}$  around a circular path of radius  $R = 30.0 \text{ m}$ . The energy lost per revolution due to synchrotron radiation is given by the empirical formula:

$$U = \frac{88.5E^4}{R}$$

where  $U$  is in keV,  $E$  is in GeV, and  $R$  is in m. Calculate the energy lost by the electron in MeV during one complete revolution.

**Solution:** We substitute the given operational parameters of the synchrotron ( $E = 7.50 \text{ GeV}$  and  $R = 30.0 \text{ m}$ ) directly into the empirical radiation formula:

$$U = \frac{88.5 \times (7.50)^4}{30.0}$$

First, compute the fourth power of the electron energy:

$$(7.50)^4 = (56.25)^2 = 3164.0625 \text{ GeV}^4$$

Substitute this value back into the numerator:

$$U = \frac{88.5 \times 3164.0625}{30.0} = \frac{280019.53125}{30.0} = 9333.984375 \text{ keV}$$

To convert the energy loss from kiloelectron-volts (keV) to megaelectron-volts (MeV), divide by  $10^3$ :

$$U(\text{in MeV}) = \frac{9333.984375 \text{ keV}}{1000 \text{ keV/MeV}} \approx 9.33 \text{ MeV}$$

## 4.3 Electromagnetism

### RL Parallel DC Circuit Transients

Two inductors  $L_1$  (inductance 1 mH, internal resistance  $3\ \Omega$ ) and  $L_2$  (inductance 2 mH, internal resistance  $4\ \Omega$ ) and a resistor  $R$  (resistance  $12\ \Omega$ ) are all connected in parallel across a 5 V battery. The circuit is switched on at time  $t = 0$ . What is the ratio of the maximum to the minimum current  $\frac{I_{\max}}{I_{\min}}$  drawn from the battery?

**Solution:** Each inductive branch can be modeled as an ideal inductance in series with its internal resistance: branch 1 ( $L_1 = 1\text{ mH}$ ,  $R_1 = 3\ \Omega$ ), branch 2 ( $L_2 = 2\text{ mH}$ ,  $R_2 = 4\ \Omega$ ), and branch 3 is purely resistive ( $R_3 = 12\ \Omega$ ). All three branches are in parallel across the DC voltage source  $V = 5\text{ V}$ .

**Step 1: Determine the Minimum Current ( $I_{\min}$ )** At the exact instant the switch is closed ( $t = 0^+$ ), the current through any physical inductor cannot change instantaneously. Thus,  $I_{L_1}(0^+) = 0$  and  $I_{L_2}(0^+) = 0$ ; the inductive branches act as open circuits. Only the pure resistive branch draws current:

$$I_{\min} = I(0^+) = \frac{V}{R_3} = \frac{5\text{ V}}{12\ \Omega} = \frac{5}{12}\text{ A}$$

**Step 2: Determine the Maximum Current ( $I_{\max}$ )** As time approaches infinity ( $t \rightarrow \infty$ ), the circuit reaches DC steady-state. The ideal inductances behave as short circuits ( $V_L = L \frac{dI}{dt} = 0$ ), leaving only their internal resistances active in parallel. The total equivalent conductance  $G_{\text{eq}}$  of the three parallel branches is:

$$G_{\text{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{3} + \frac{1}{4} + \frac{1}{12}$$

Finding a common denominator of 12:

$$G_{\text{eq}} = \frac{4 + 3 + 1}{12} = \frac{8}{12} = \frac{2}{3}\text{ S}$$

The equivalent steady-state resistance  $R_{\text{eq}}$  is the reciprocal of conductance:

$$R_{\text{eq}} = \frac{1}{G_{\text{eq}}} = \frac{3}{2}\ \Omega = 1.5\ \Omega$$

The maximum current drawn from the battery occurs at steady-state:

$$I_{\max} = I(\infty) = \frac{V}{R_{\text{eq}}} = \frac{5\text{ V}}{3/2\ \Omega} = \frac{10}{3}\text{ A}$$

**Step 3: Calculate the Current Ratio** Dividing the maximum steady-state current by the initial minimum current:

$$\frac{I_{\max}}{I_{\min}} = \frac{10/3}{5/12} = \frac{10}{3} \times \frac{12}{5} = 2 \times 4 = 8$$